

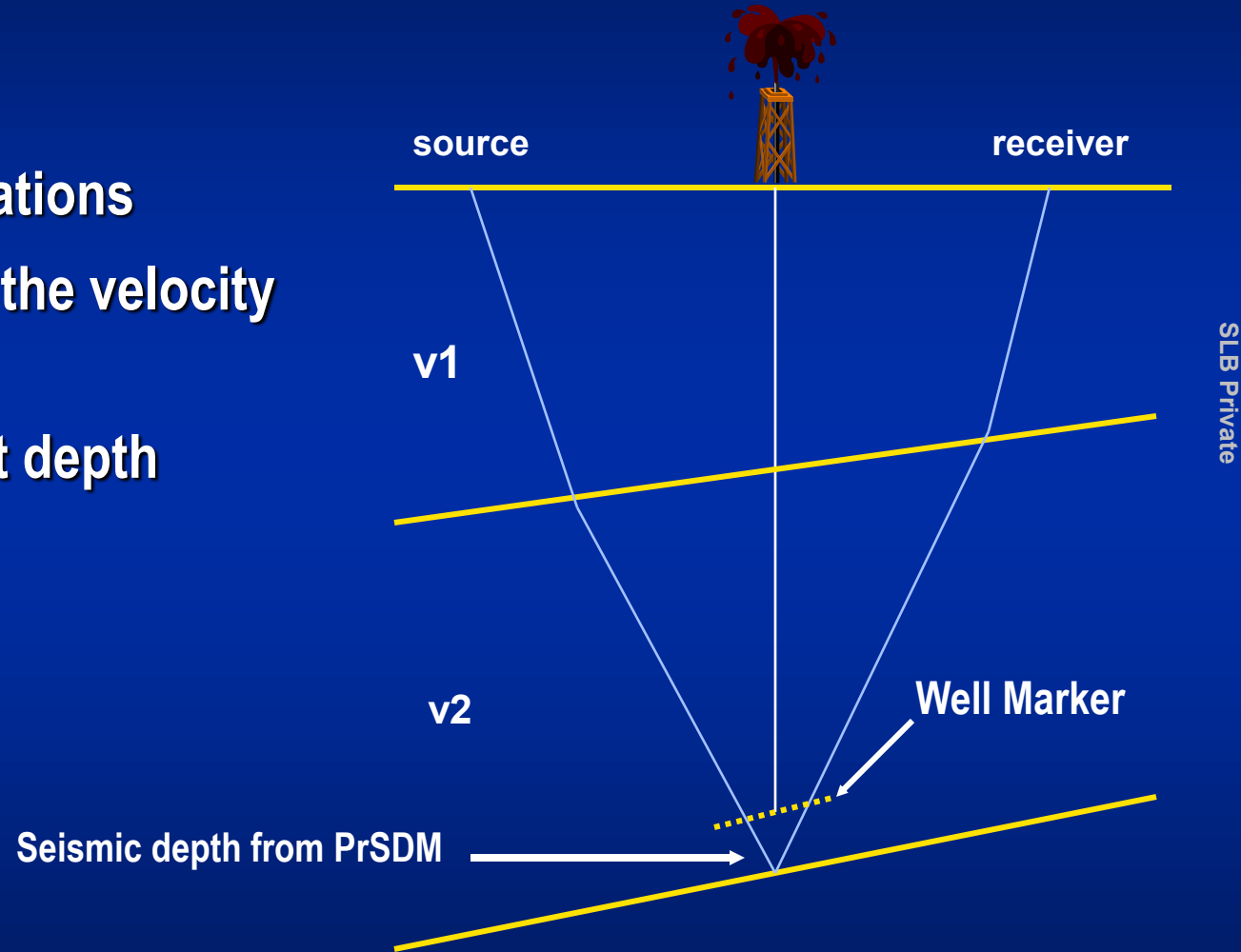
Introduction to Anisotropy

Outline

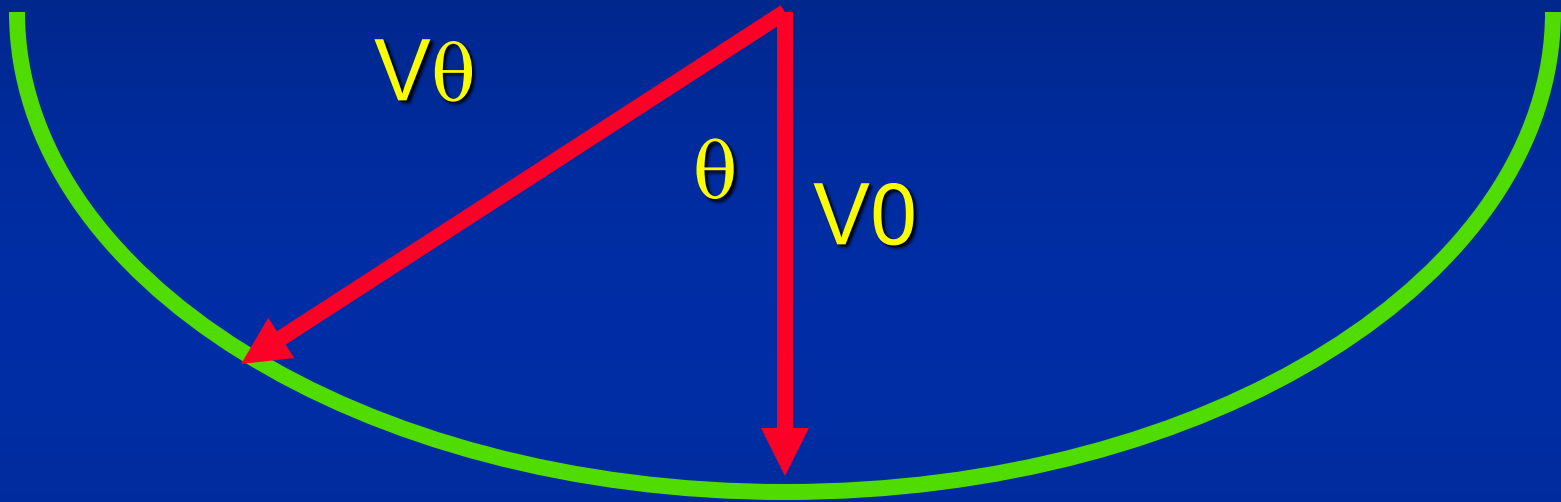
- *Anisotropy, what is it and what causes it?*
- Types of anisotropy
- Thomsen's parameters, epsilon and delta defined and explained
- Symptoms of anisotropy
- Determining anisotropic parameters and some problems
- Updating anisotropic parameters, a possible workflow

Why seismic depths do not tie the well depths

- **Anisotropy**
- Lateral velocity variations
- Parameterisation of the velocity model
- Lack of resolution at depth



Anisotropy



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Velocity varies with propagation direction

Thomsen, Leon, 1986. 'Weak elastic anisotropy' Geophysics, 51,
1954-1966

Some causes of anisotropy

- Layering
 - Beds thinner than the seismic wavelength with differing velocities
- Fracturing
 - Can rapidly become complex
- Intrinsic
 - The crystal exhibit different behaviour in different directions
 - Can be very complex

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Types of anisotropy

- TI – transverse isotropy, symmetrical about a single axis
 - VTI – vertical transverse isotropy
 - HTI – horizontal transverse isotropy
 - TTI – tilted transverse isotropy
- More general and most realistic
 - Orthorhombic, 3 orthogonal axes, 9 parameters to describe it

Isotropic wave propagation

K is bulk modulus

μ is shear modulus (stress/strain)

ρ is density

$$v_p = \left(\frac{K + 4/3\mu}{\rho} \right)^{1/2}$$

$$v_s = \left(\frac{\mu}{\rho} \right)^{1/2}$$

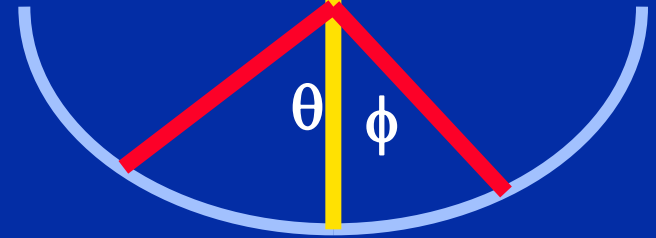


Spherical wavefront

VTI

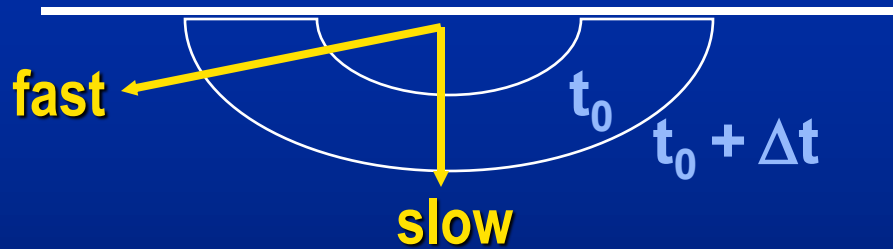
Axis of symmetry

Axis vertical

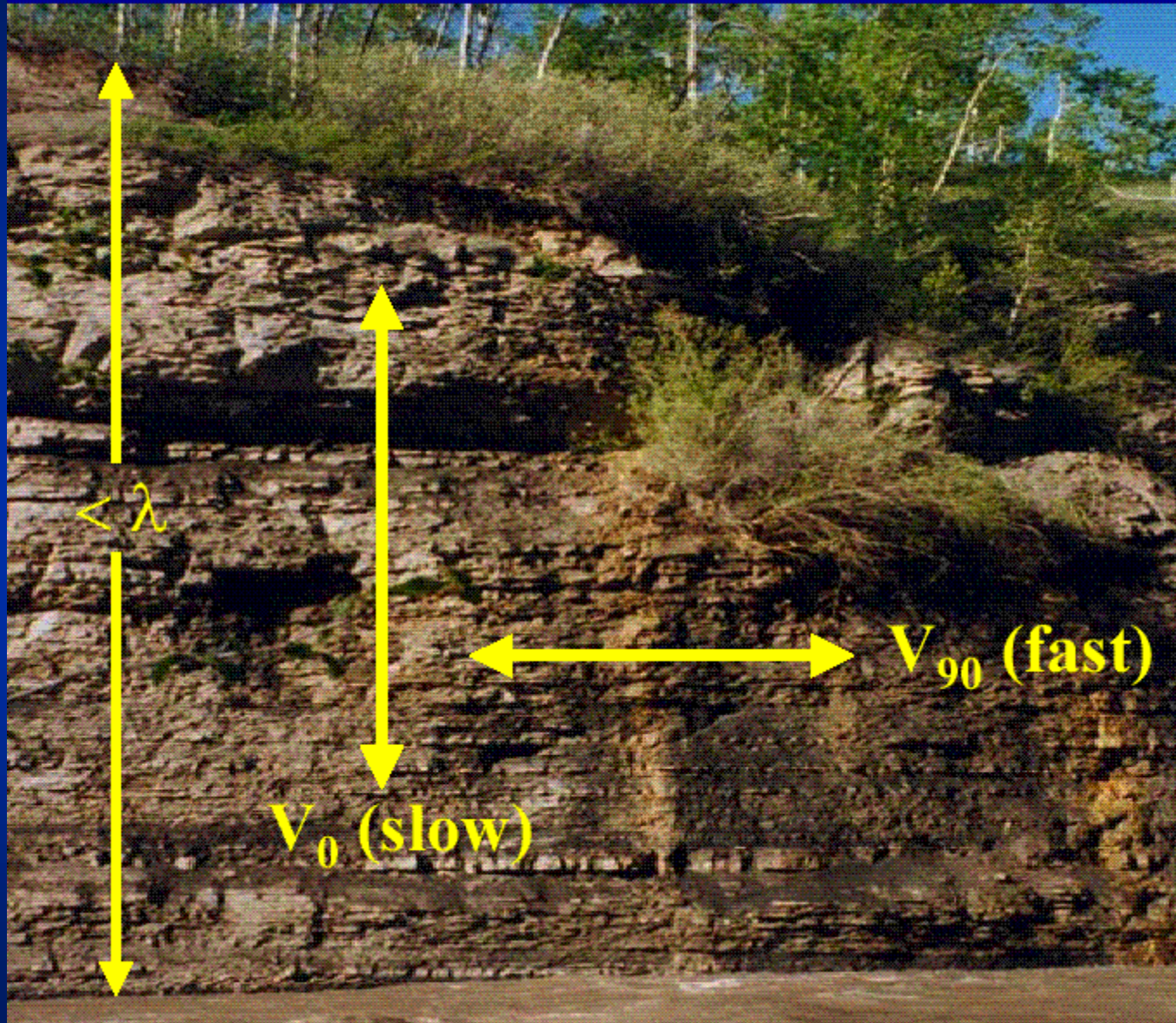


$\theta = \phi \Rightarrow V_{\theta} = V_{\phi}$ for all azimuth

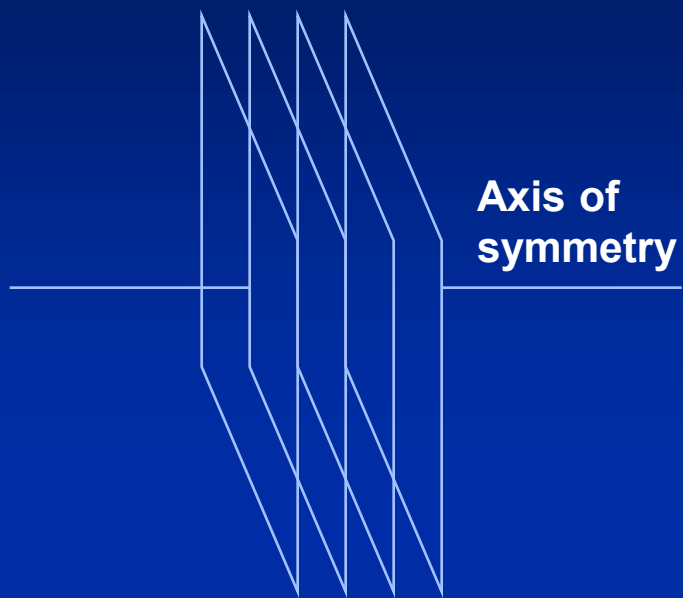
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VTI in Practice



HTI

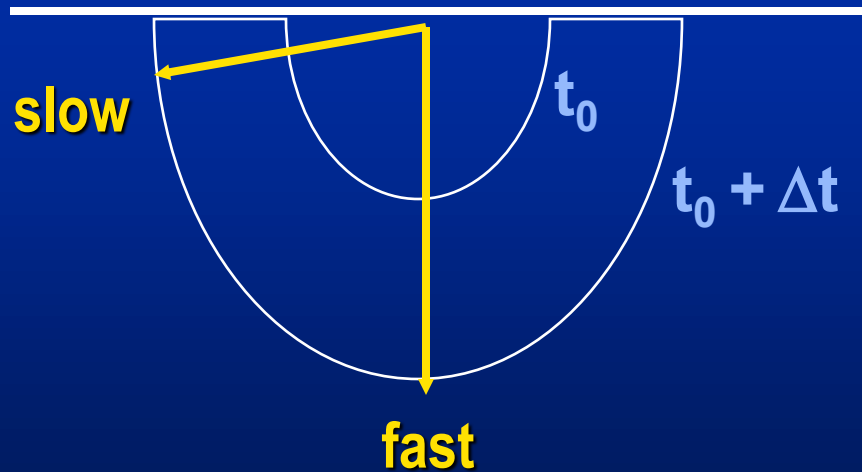


Axis horizontal



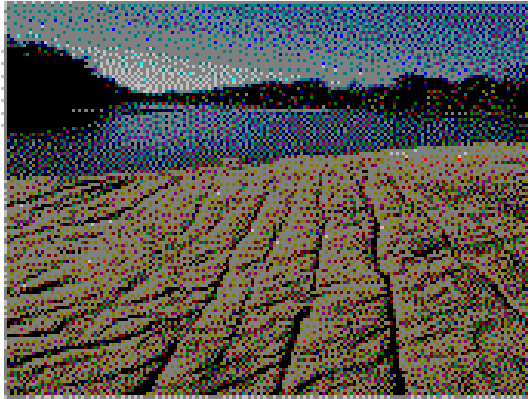
$$\theta = \phi \Rightarrow V_{\theta} = V_{\phi}$$

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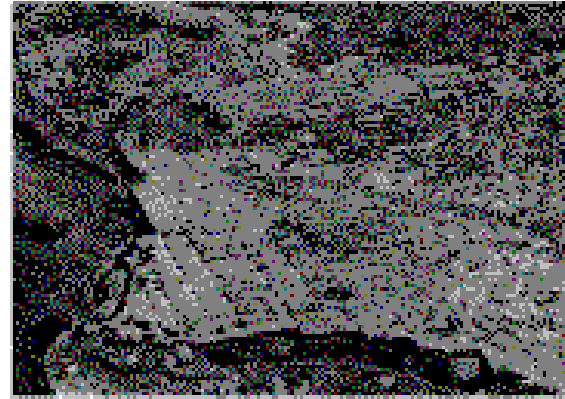
HTI in Practice

SANDSTONE



<http://pyrite.igs.indiana.edu/indgeol/reference/>

SANDSTONE

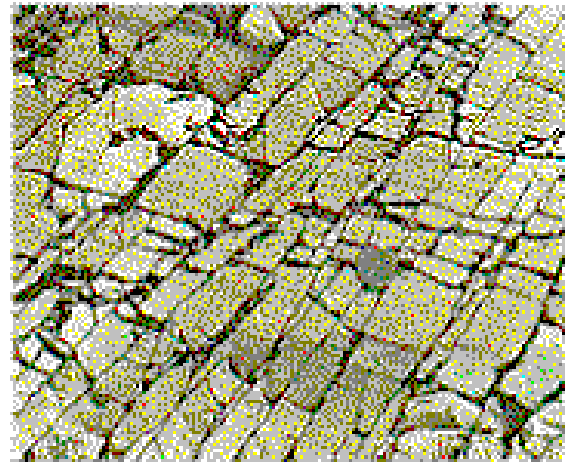


(J.Olson)

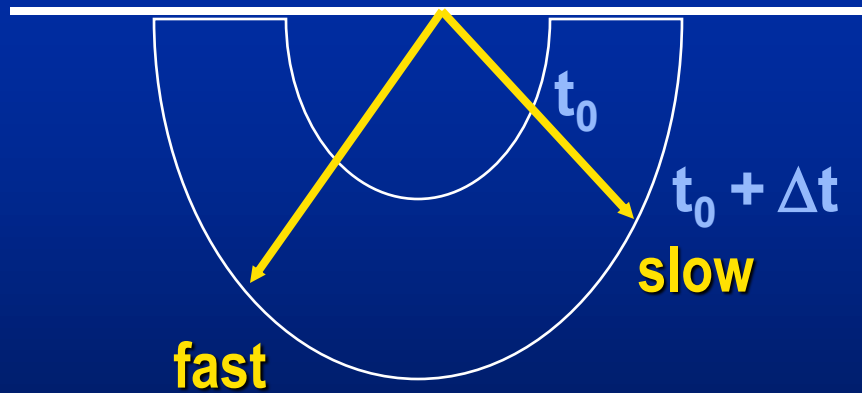
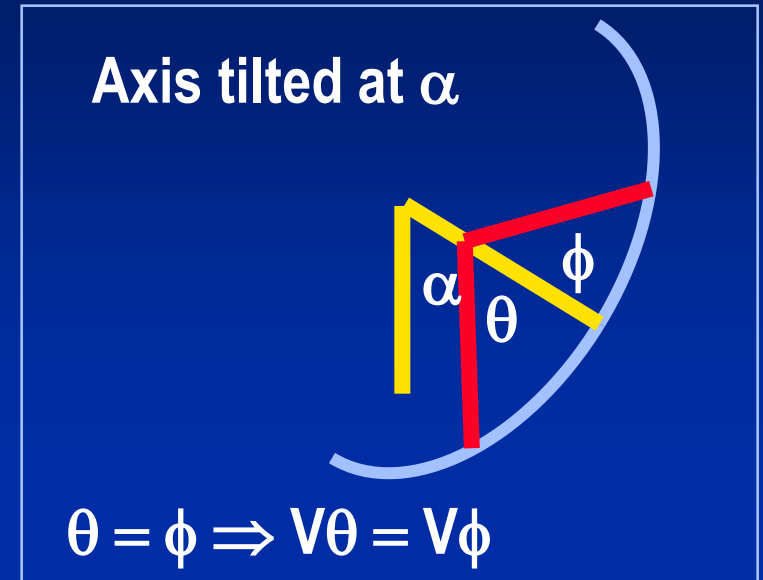
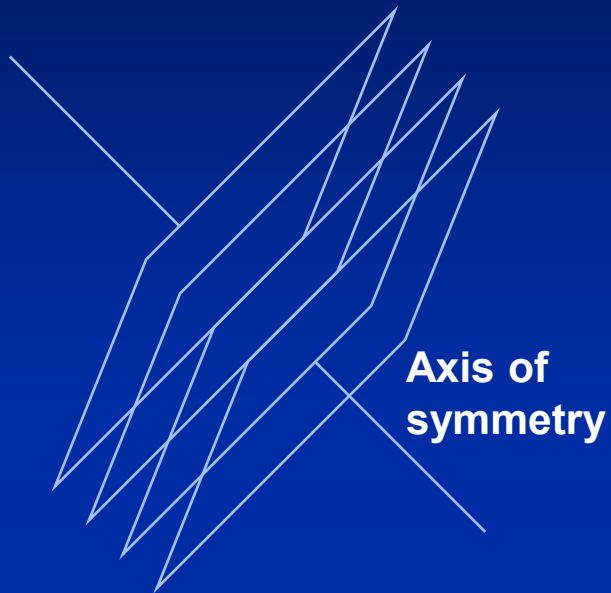
SANDSTONE



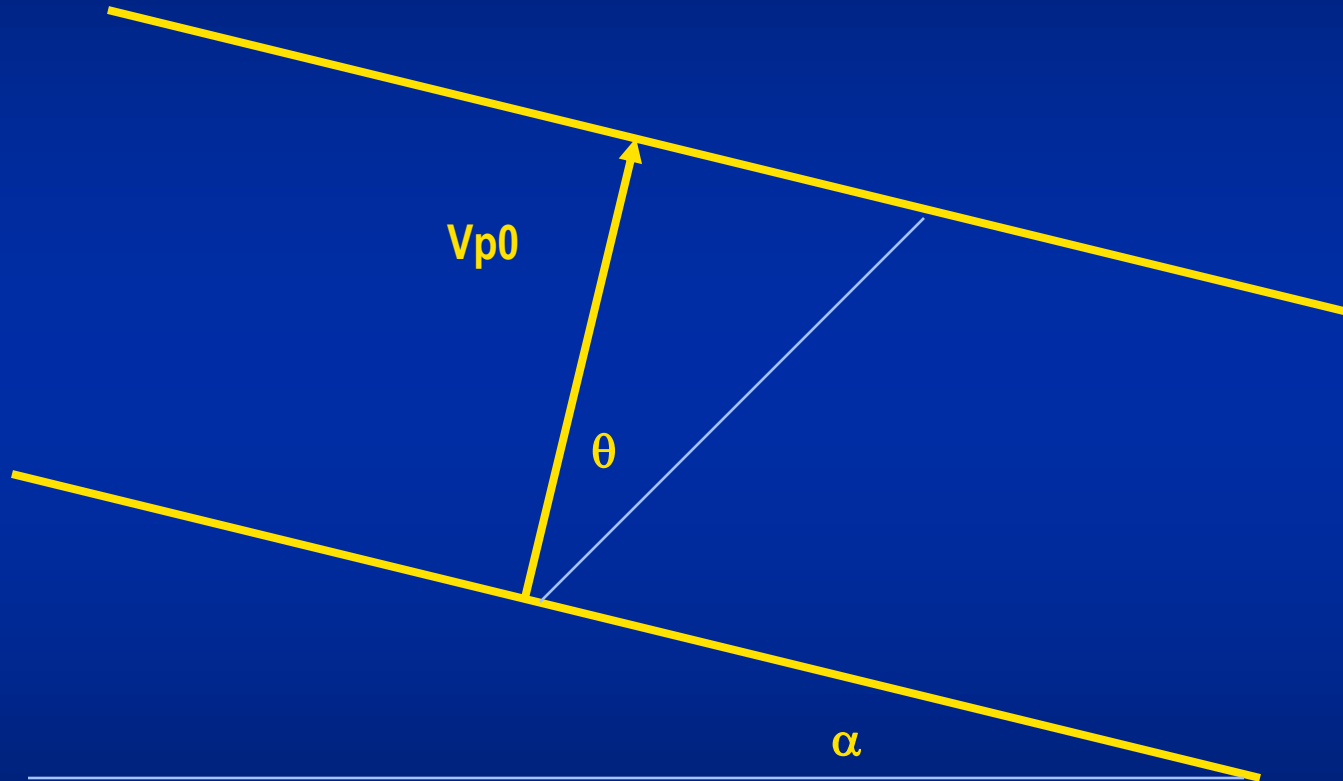
LIMESTONE



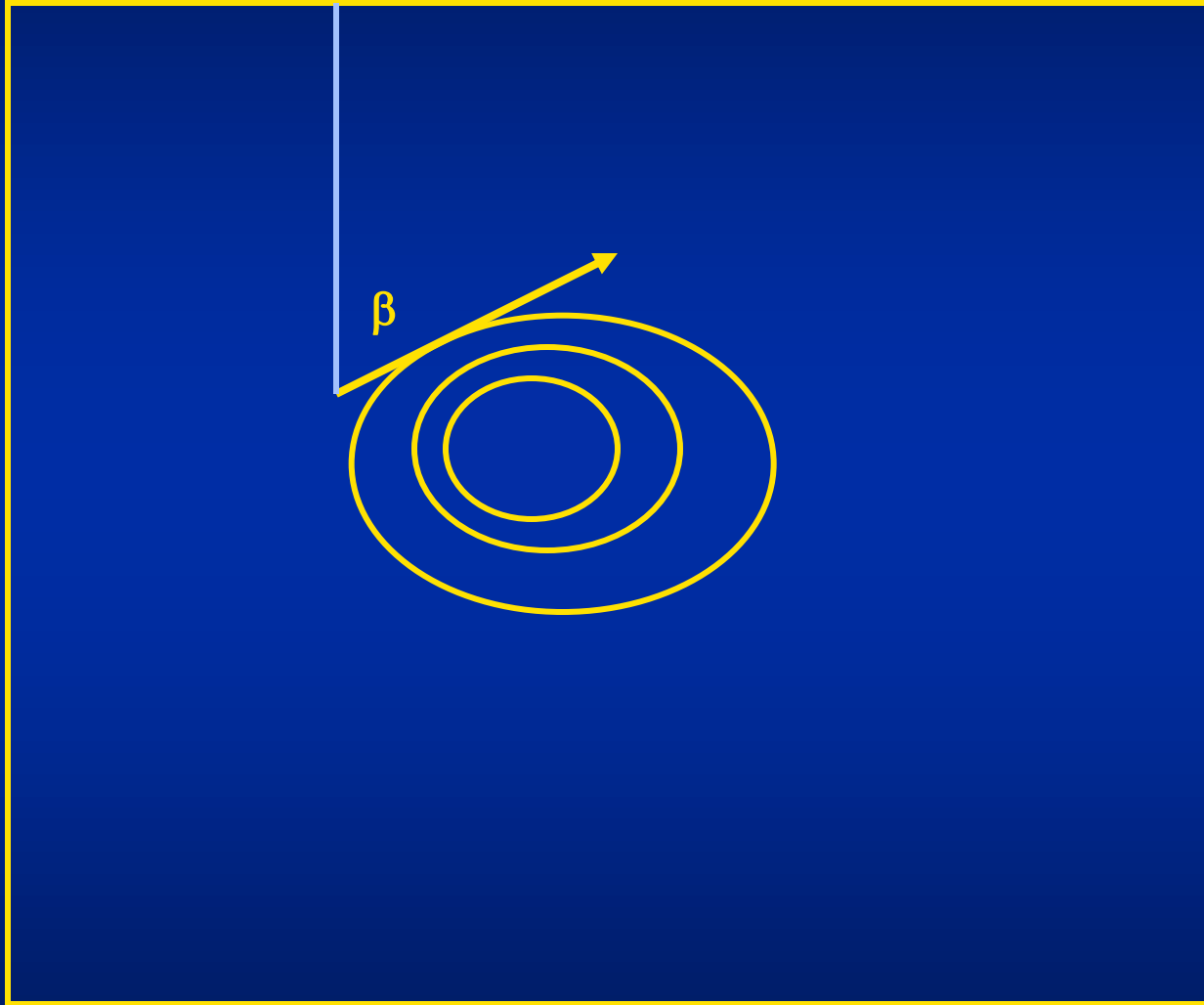
(Roberts 1989)



TTI – dip angle α 0 to 90°



TTI – azimuth angle β 0 to 360°



TTI in Practice

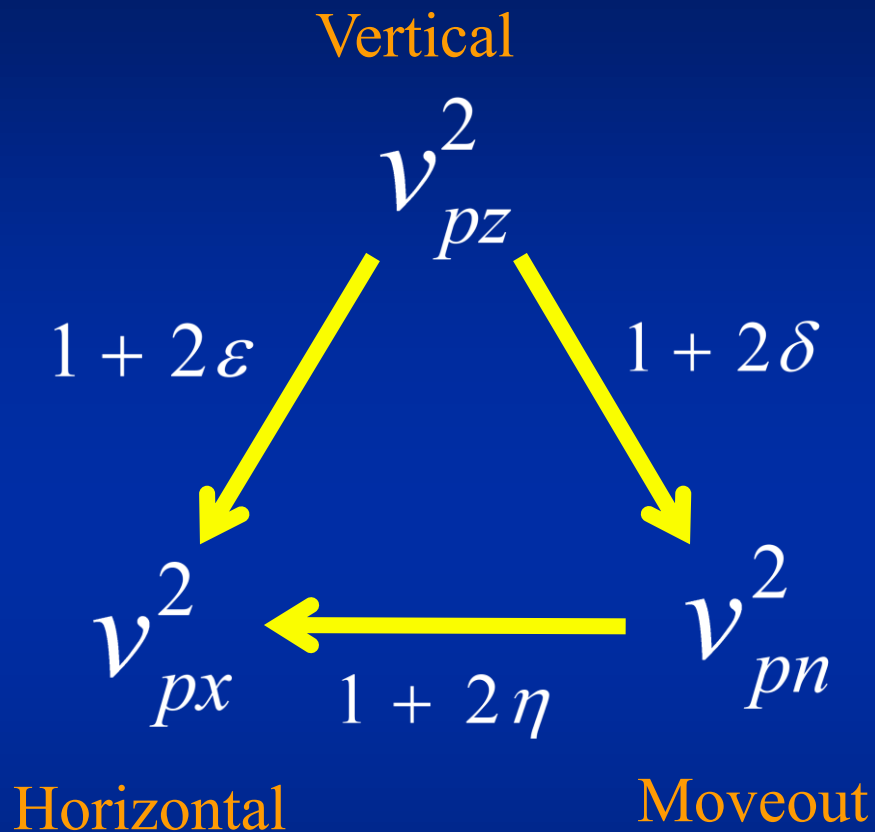


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VTI Parameterization – Remember This!



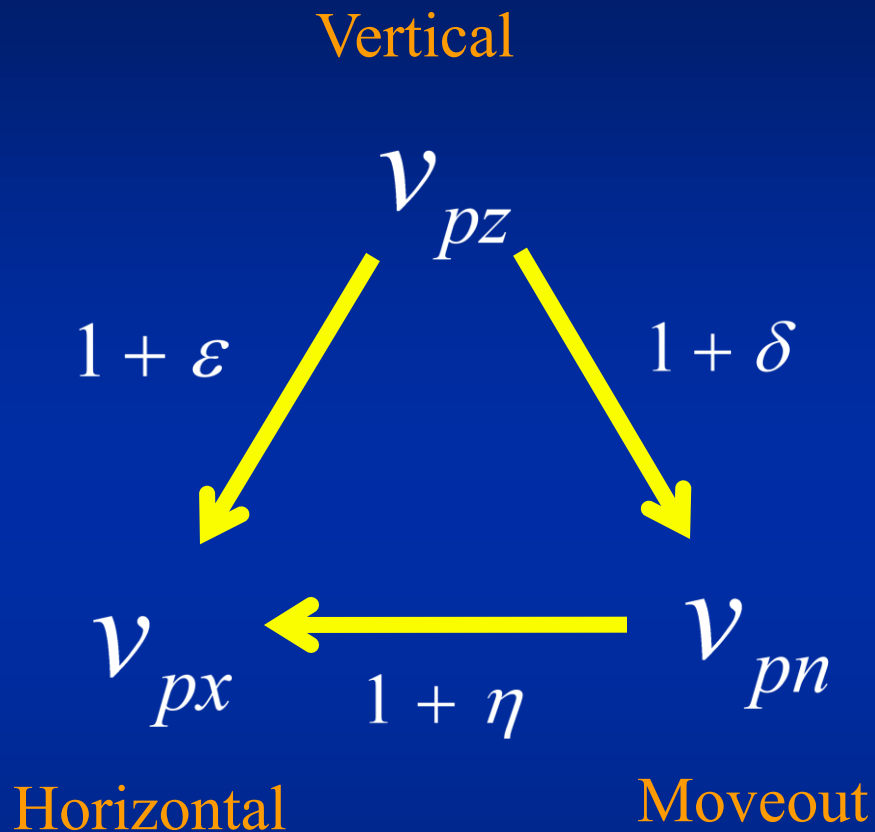
$$v_{px}^2 = v_{pz}^2 (1 + 2\varepsilon)$$

$$v_{pn}^2 = v_{pz}^2 (1 + 2\delta)$$

$$v_{px}^2 = v_{pn}^2 (1 + 2\eta)$$

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VTI Parameterisation (Approximate)



$$\frac{v_{px}}{v_{pz}} \approx 1 + \varepsilon$$

$$\frac{v_{pn}}{v_{pz}} \approx 1 + \delta$$

$$\frac{v_{px}}{v_{pn}} \approx 1 + \eta$$

$$\eta \approx \varepsilon - \delta$$

What do epsilon and delta mean physically?

Epsilon is the fractional difference between the vertical and horizontal velocities

$$\varepsilon = (V_p(90) - V_p(0)) / V_p(0)$$

Delta is not physically very meaningful, but it is a function of the vertical velocity and the moveout velocity and is roughly related to the percentage well mistie for flat layers.

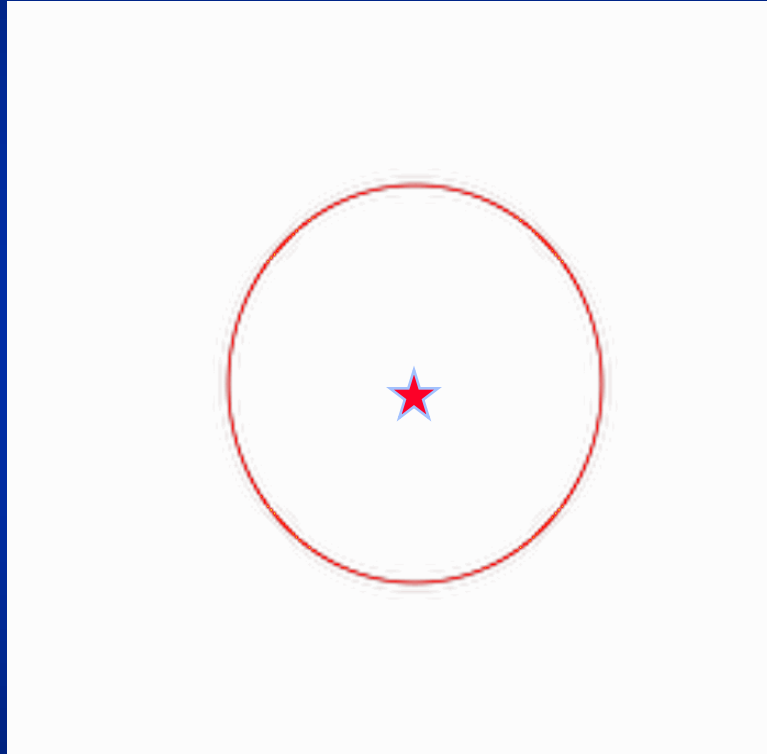
$$V_{pnmo} = V_p(0) (1 + \delta)$$

$$\delta \Rightarrow (Z_{seismic} - Z_{well}) / Z_{seismic}$$

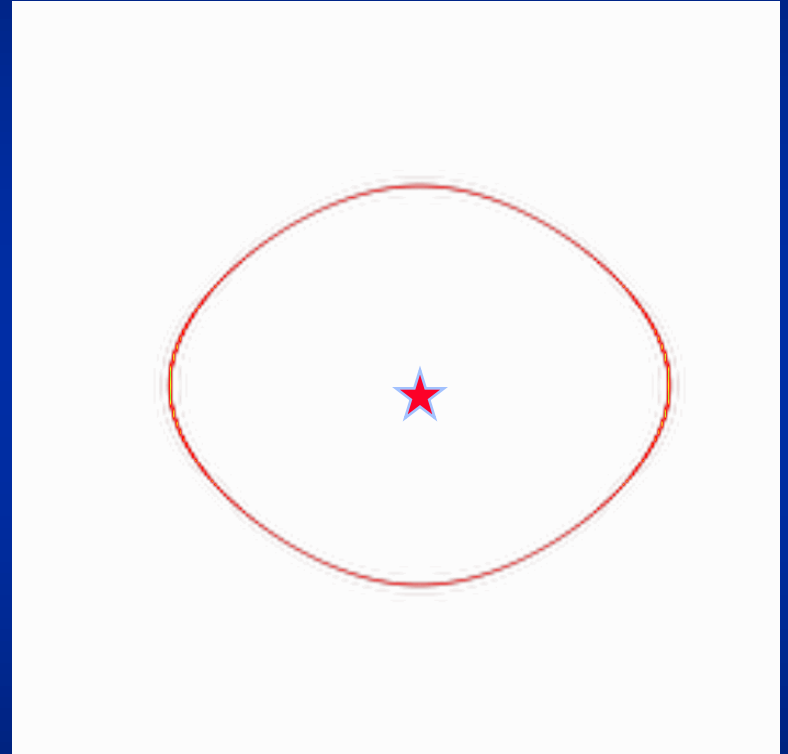
What is seismic anisotropy?

$x \rightarrow$

$z \downarrow$



Isotropic



Anisotropic

(P-wave only)
Schlumberger-Private

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VTI: ranges of parameters

Anisotropy of Rock Samples
(average values from Z. Wang, 2002)

Rock type	ϵ	δ	γ	σ	η
Shales	0.21	0.03	0.20	0.55	0.19
Sands	0.07	0.05	0.04	0.04	0.02
Carbonates	0.03	0.01	0.01	0.05	0.03

Carbonates and clay-free sands
are almost isotropic

Outline

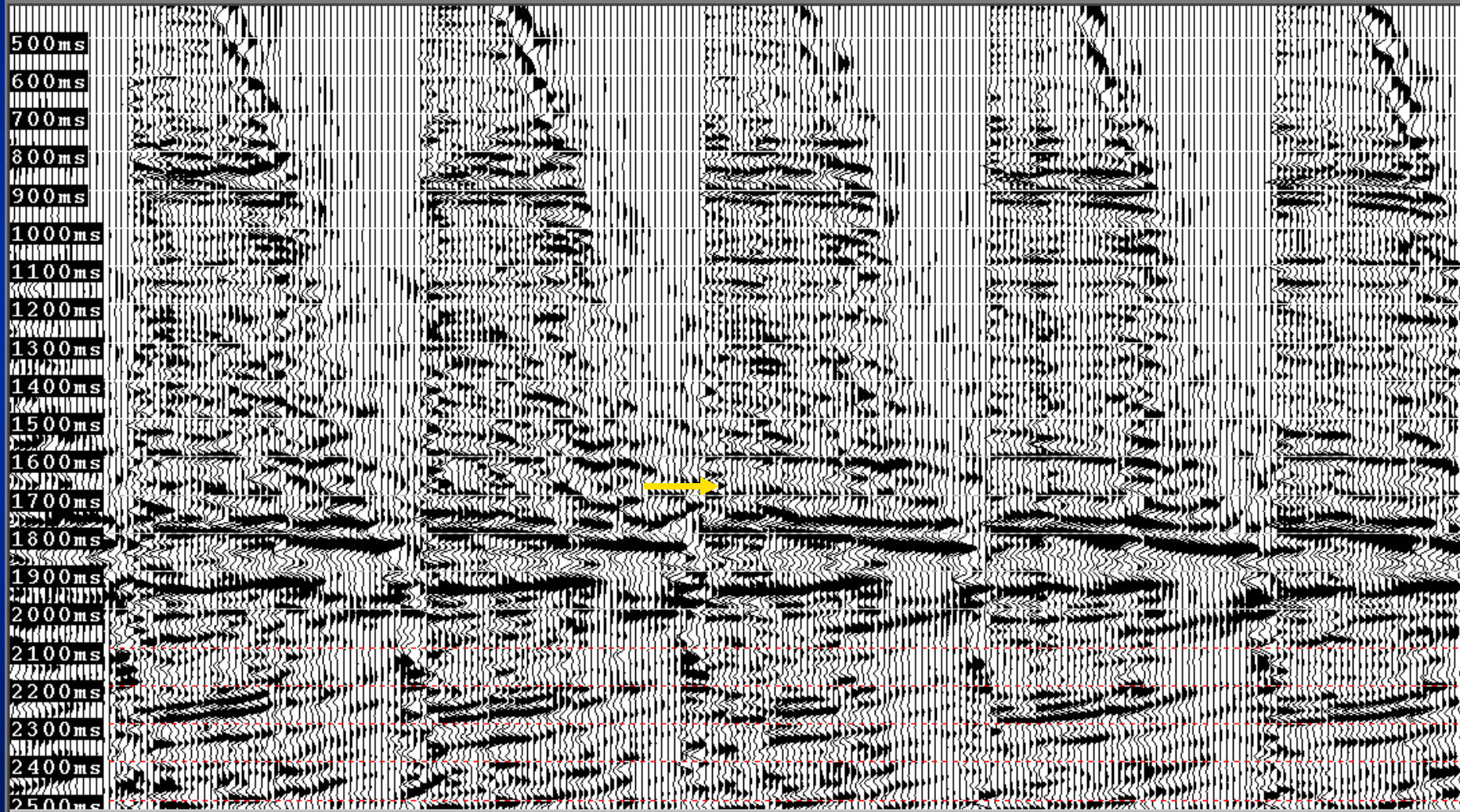
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Symptoms of anisotropy

- Well misties
- 'Hockey stick' effects
- Inconsistent imaging of dipping events
- Velocity variations correlating with structure
- Problems with imaging different dips

Southern North Sea depth gathers

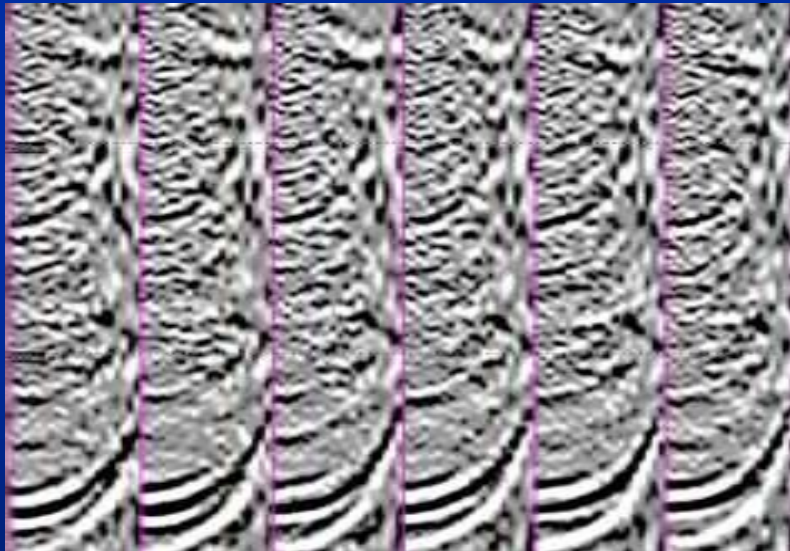
Well depth = 1644m



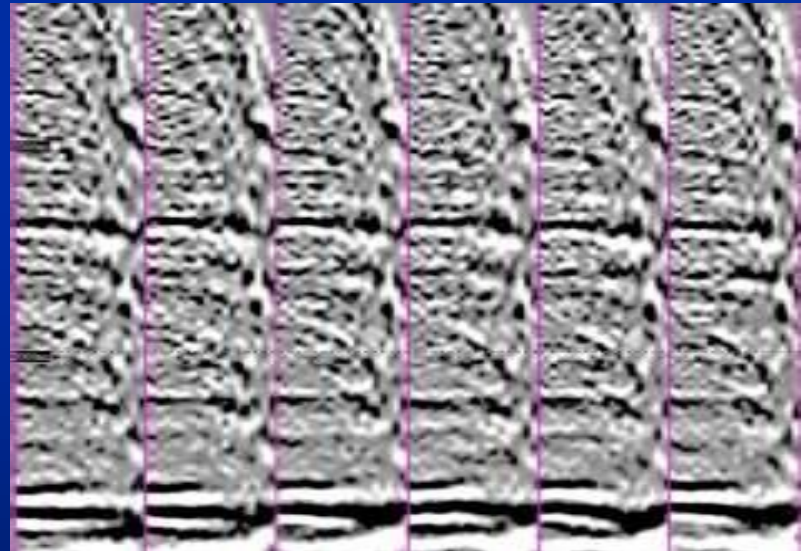
What does this look like on a migrated gather?

If we depth migrate with the correct vertical velocity but without anisotropy, then the depth migrated gathers will not be flat - here is an example:

Isotropic Depth



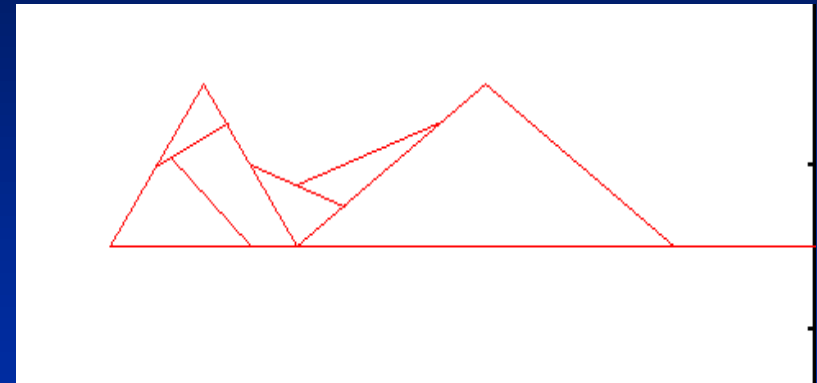
Anisotropic Depth



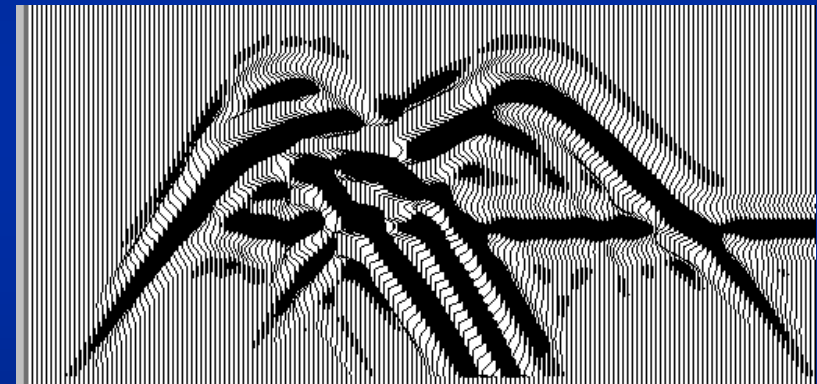
same v_0 velocity field, offsets increase to right

Conflicting dips

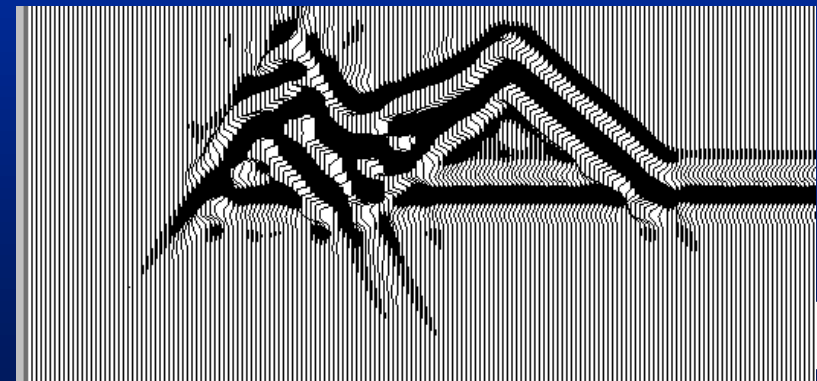
Model



Optimum isotropic
imaging



Anisotropic
imaging



Solution

Include anisotropy, but this requires extra control during the velocity model building phase, for example:

- **Well information**
- **Understanding geology / geophysics**
- **VSP information**

Anisotropic parameters for depth imaging

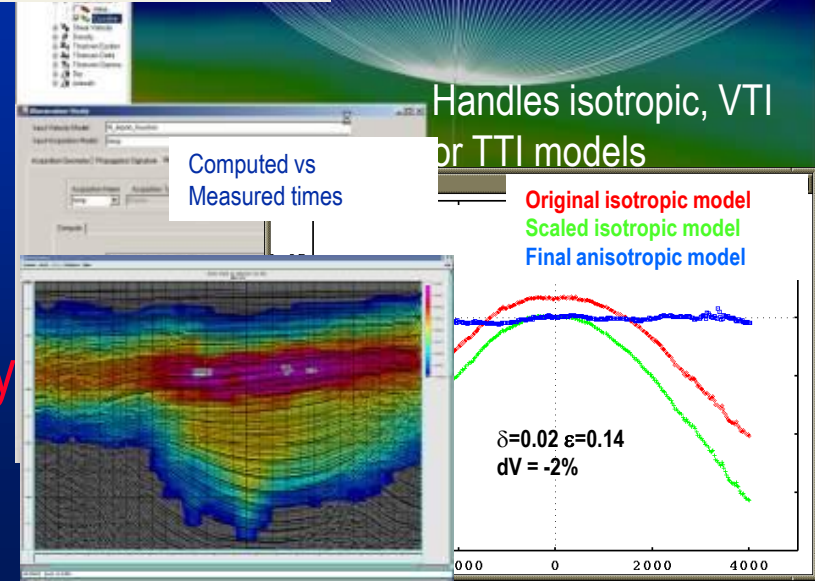
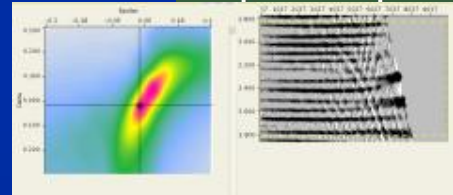
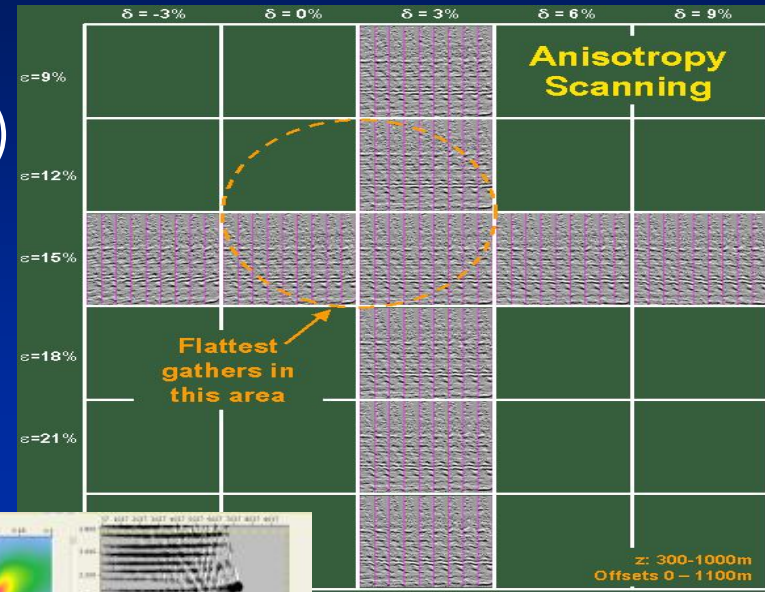
- Time imaging
 - Effective parameters (η)
 - Local effects
 - Best stacking
- Depth imaging
 - Increased significance of anisotropic parameters
 - Parameters affect the propagation of rays
 - Deeper data cannot “ignore” overburden effects
- Parameters should be more closely related to subsurface properties

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How do we derive anisotropic parameters?

1. Compute (approximate) delta from isotropic well mis-tie (or V_{pz}/V_{nmo} ratio)
2. Use regional values (Lithology knowledge, assumptions e.g. compaction)
3. Migration scans for δ & ϵ
4. Forward Modeling 1D or 3D
 - Using pre-stack seismic data
 - Walkaway VSPs
5. (localized) Constrained Tomographic updates
6. 1D VTI Inversion (& V_{pn}/V_x analysis)
7. Geomechanical modelling
8. Other measurements (sonic & resistivity scanners) ???

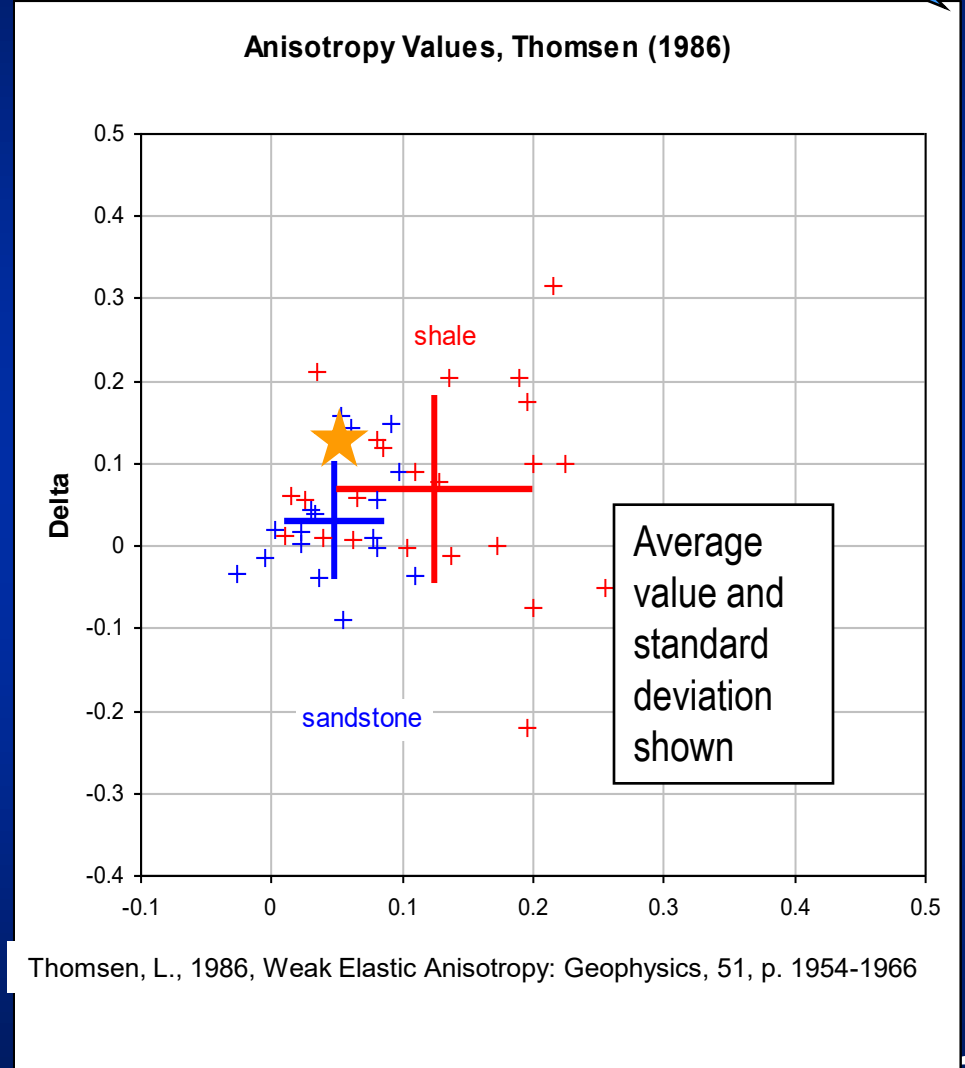


Method 1: Computing delta from well misties

- Can **roughly** (valid for small δ and low dip only) estimate δ from the well mistie
 - $\delta = (Z - Z_{\text{well}})/(Z)$
 - Use this **ONLY** as a background information about its magnitude
 - Danger is that long offset term also gets included
 - always check, confirm with **OTHER** methods

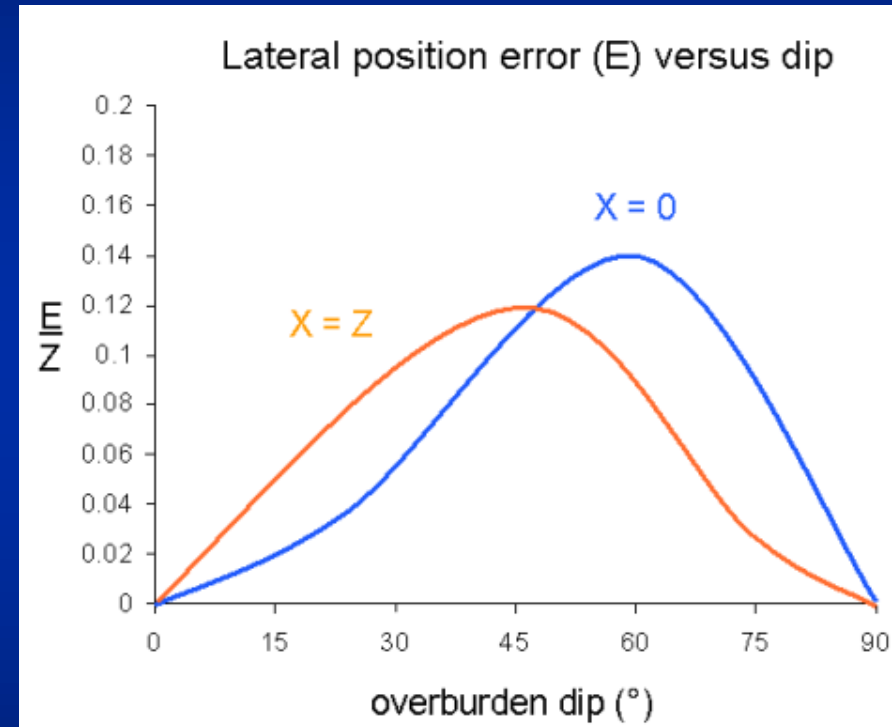
Method 2: Use of Regional Values - Constant Anisotropy Approximation

- A technique successfully used in Canadian foothills and South America
- In the Canadian Foothills setting the Thomsen anisotropy parameters $\epsilon = 12\%$ and $\delta = 4\%$ often (but not always) gives reasonable results.
 - These values fall between average sand and shale anisotropy values and is reasonable because the clastic sequence in this region is a mixture of sands and shales.



Method 3: Migration Scans for δ & ε

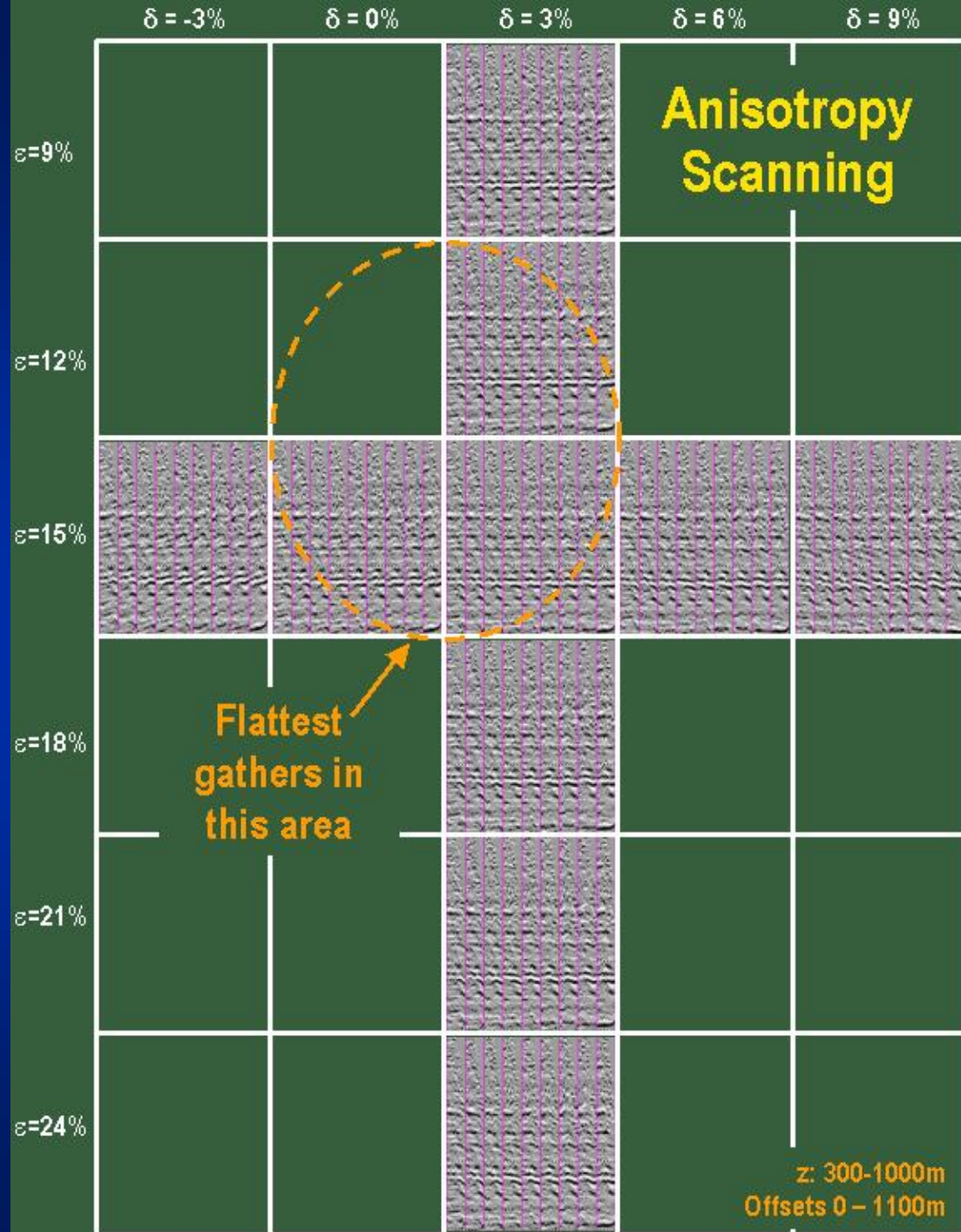
- This technique can be used to derive the effective 'average' anisotropy field or to derive values for different layers
 - Latter implies layer stripping
- When evaluating the results and making choices evaluate the following:
 - Flatness of gathers
 - especially at far offsets
 - Good to use for relatively simple geology
 - Stack improvements
 - especially dipping reflector continuity
 - Must to use in complex geology
 - Minimization of sideslip
 - Visible on stacks
 - Observed as less difference between different offsets or partial stacks



Vestrum, R., pers. comm.

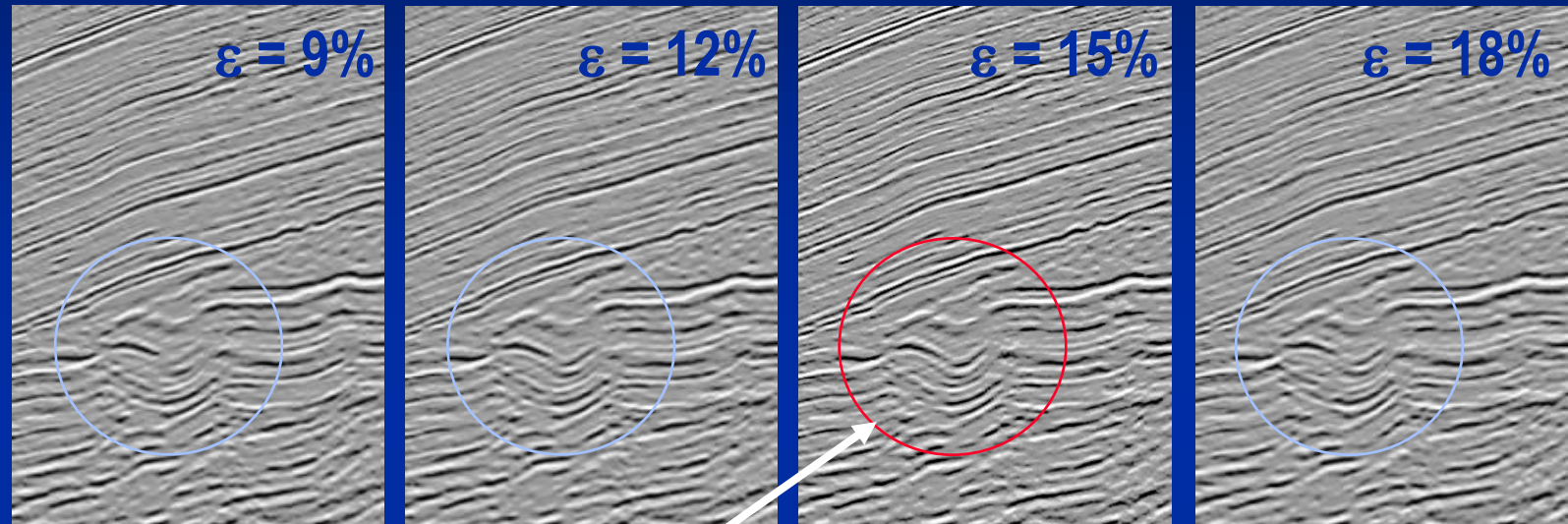
Method 3: Migration Scans for δ & ε

- Studying the flatness of events on CIP gathers migrated with different anisotropy



Method 3: Migration Scans for δ & ε

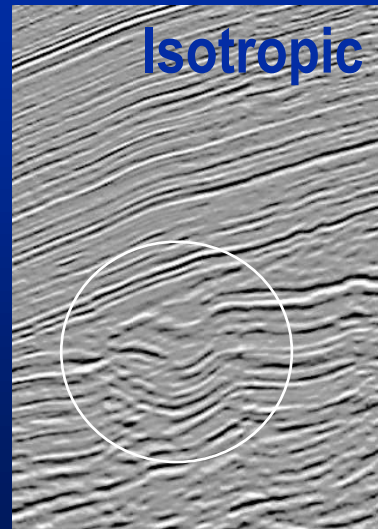
Evaluating different stacks after ε scans



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Studying the effect on images as well

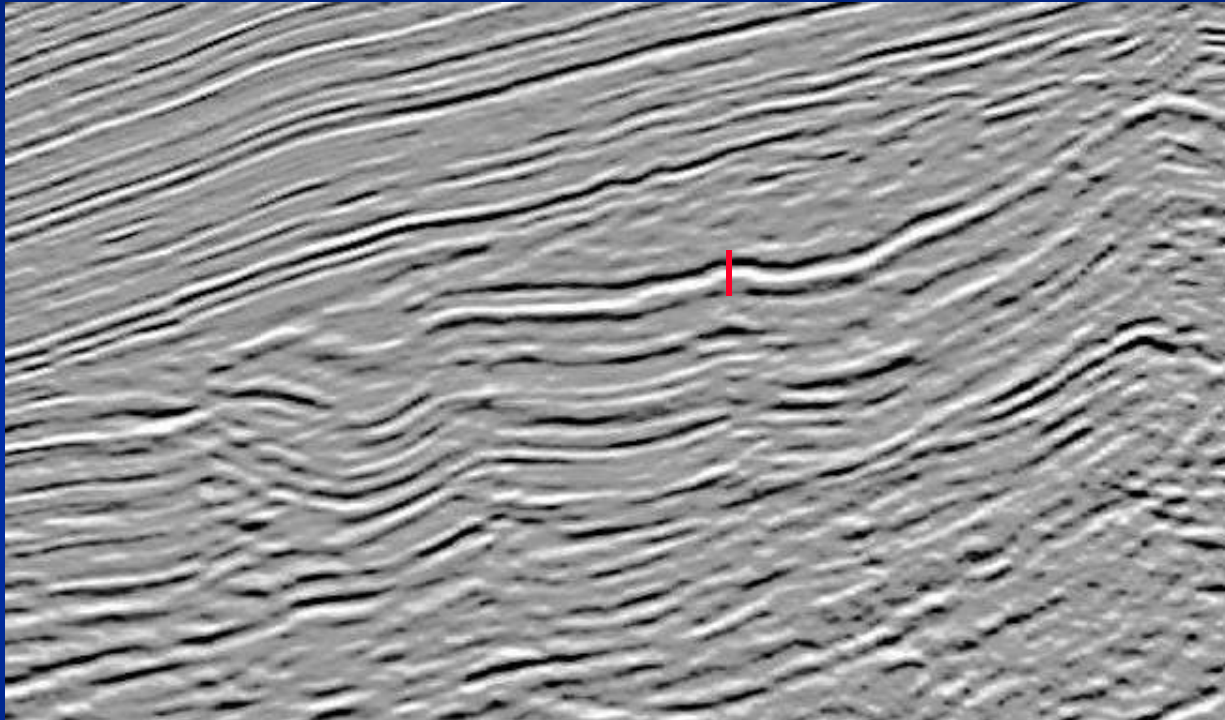
Best focusing



Method 3: Migration Scans for δ & ε

In more detail ...focusing & sideslip

Isotropic
(corrected depth)

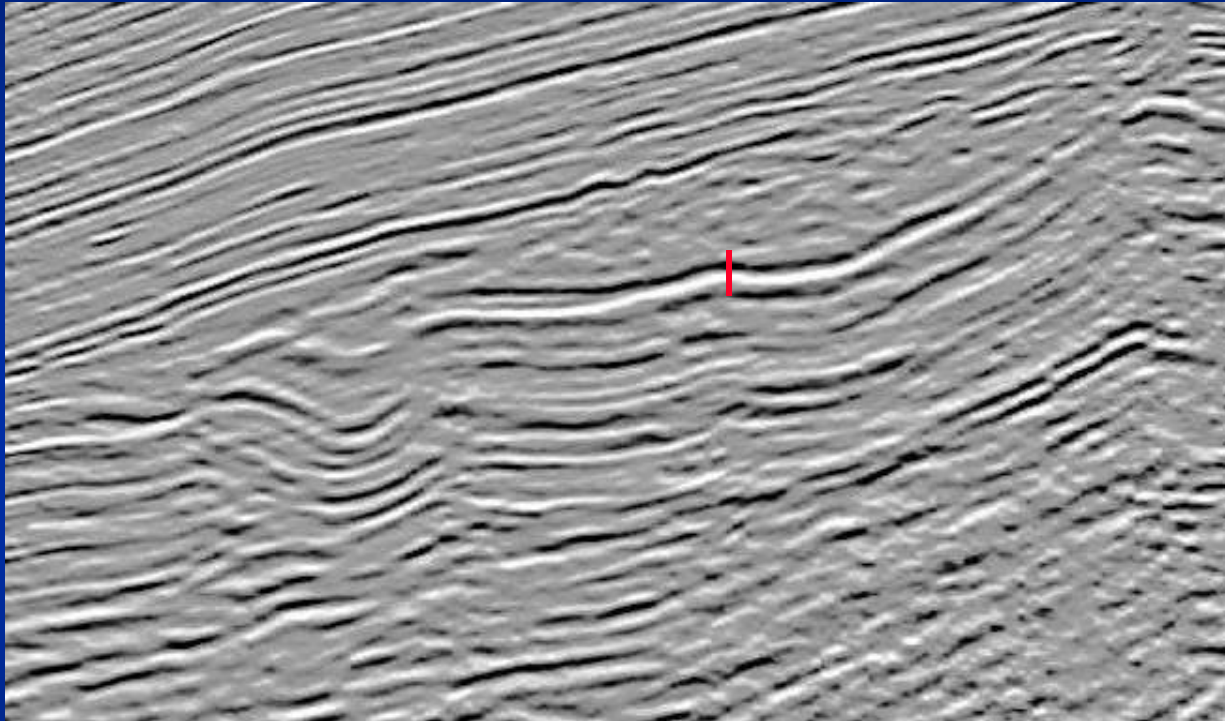


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Method 3: Migration Scans for δ & ε

In more detail ...focusing & sideslip

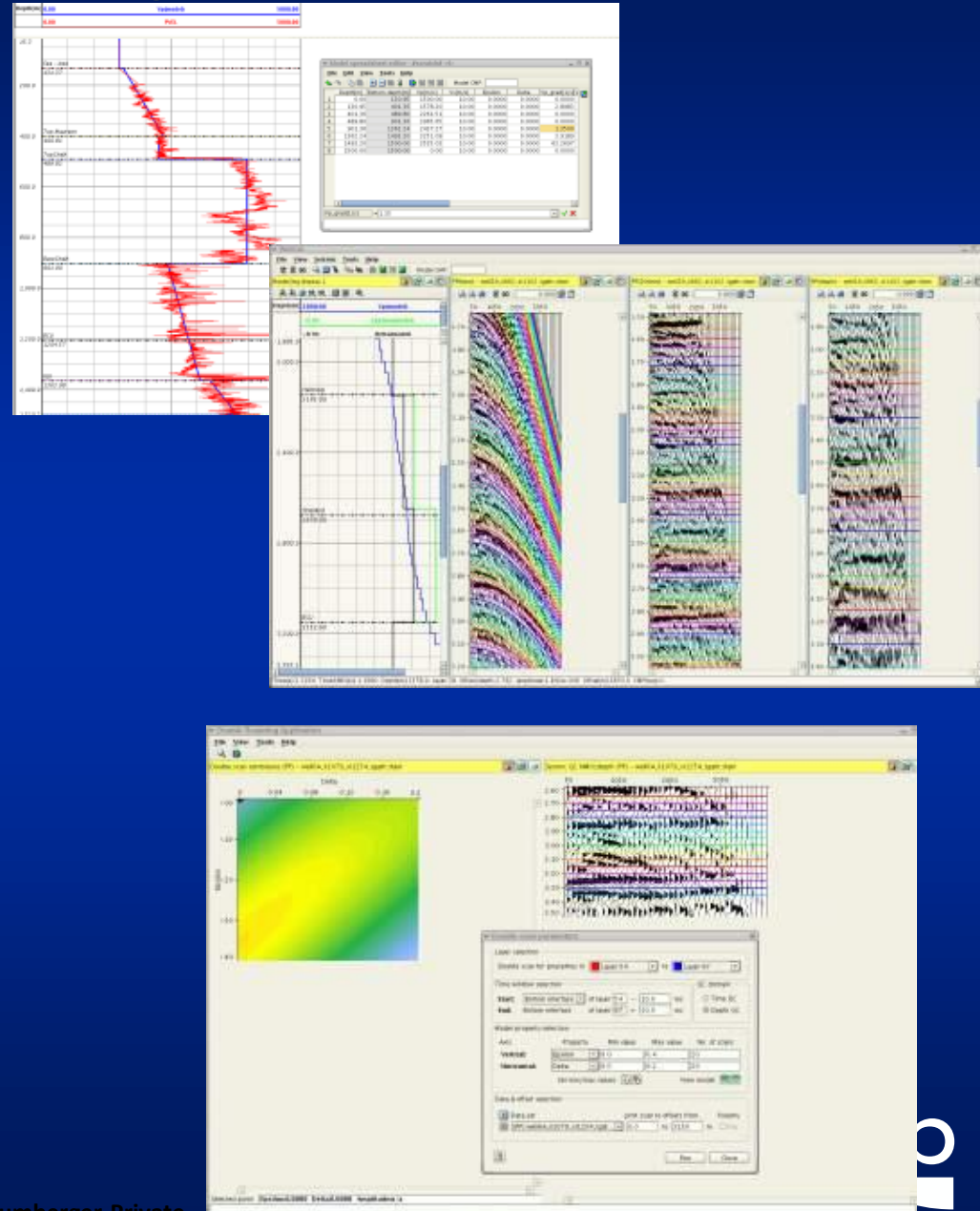
Anisotropic - ε 15 δ 3



As expected, structure moves down dip and is better focused after anisotropic depth migration

Method 4: Forward Modeling - SeisCal

- Anisotropy estimation and validation
- Useful for “what if” scenarios
 - how sensitive is data to velocity or anisotropy changes in a thin layer
- PS capable
- Gradient estimation and analysis (layered and initial models).
- Comparison of model velocities with sonics



Limitations: 1D velocity model and only VTI ray-tracing

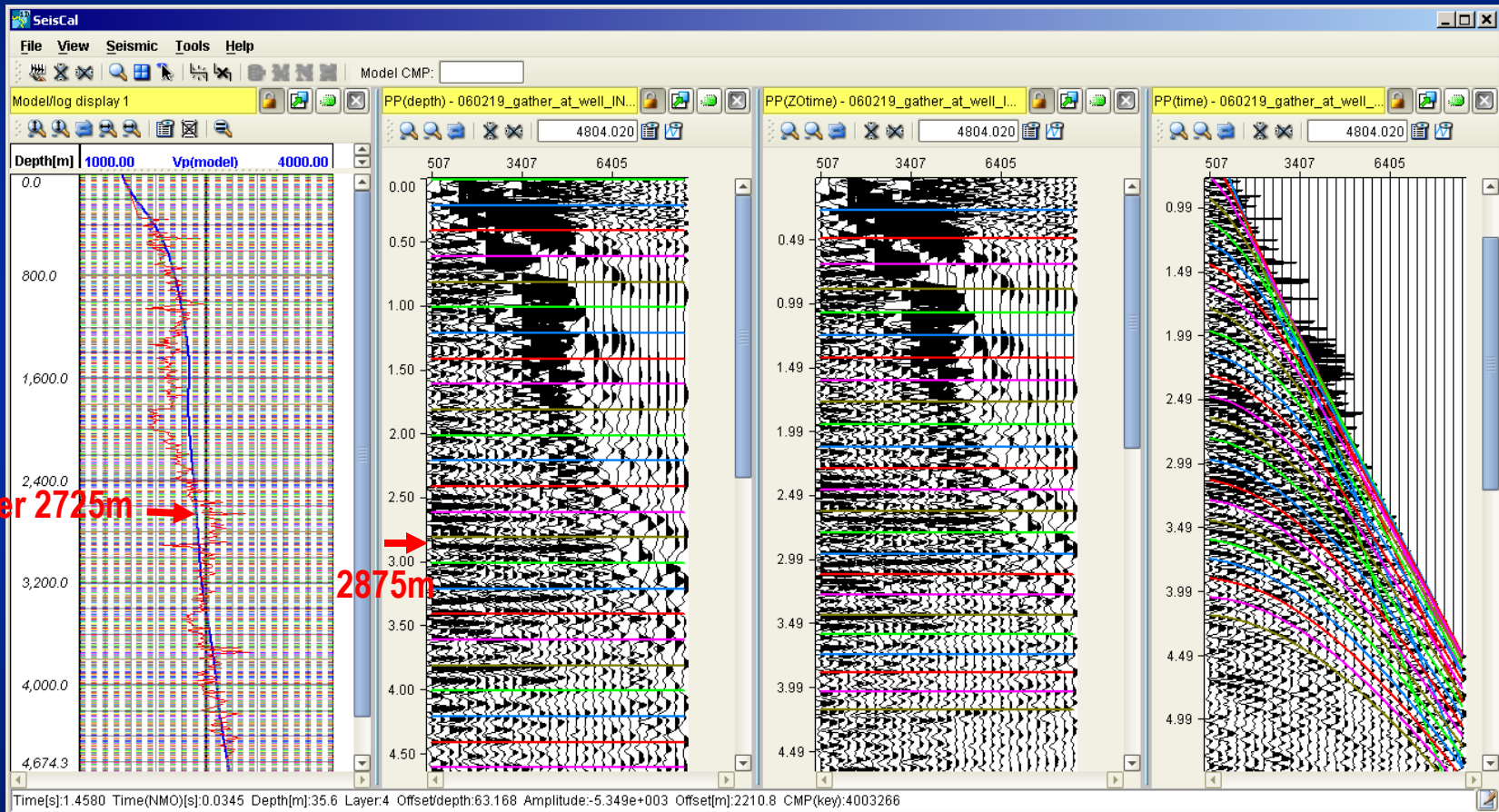
SeisCal forward modelling - Isotropic PSDM model

Log vel / Vv model

converted depth

converted time

Input time gather



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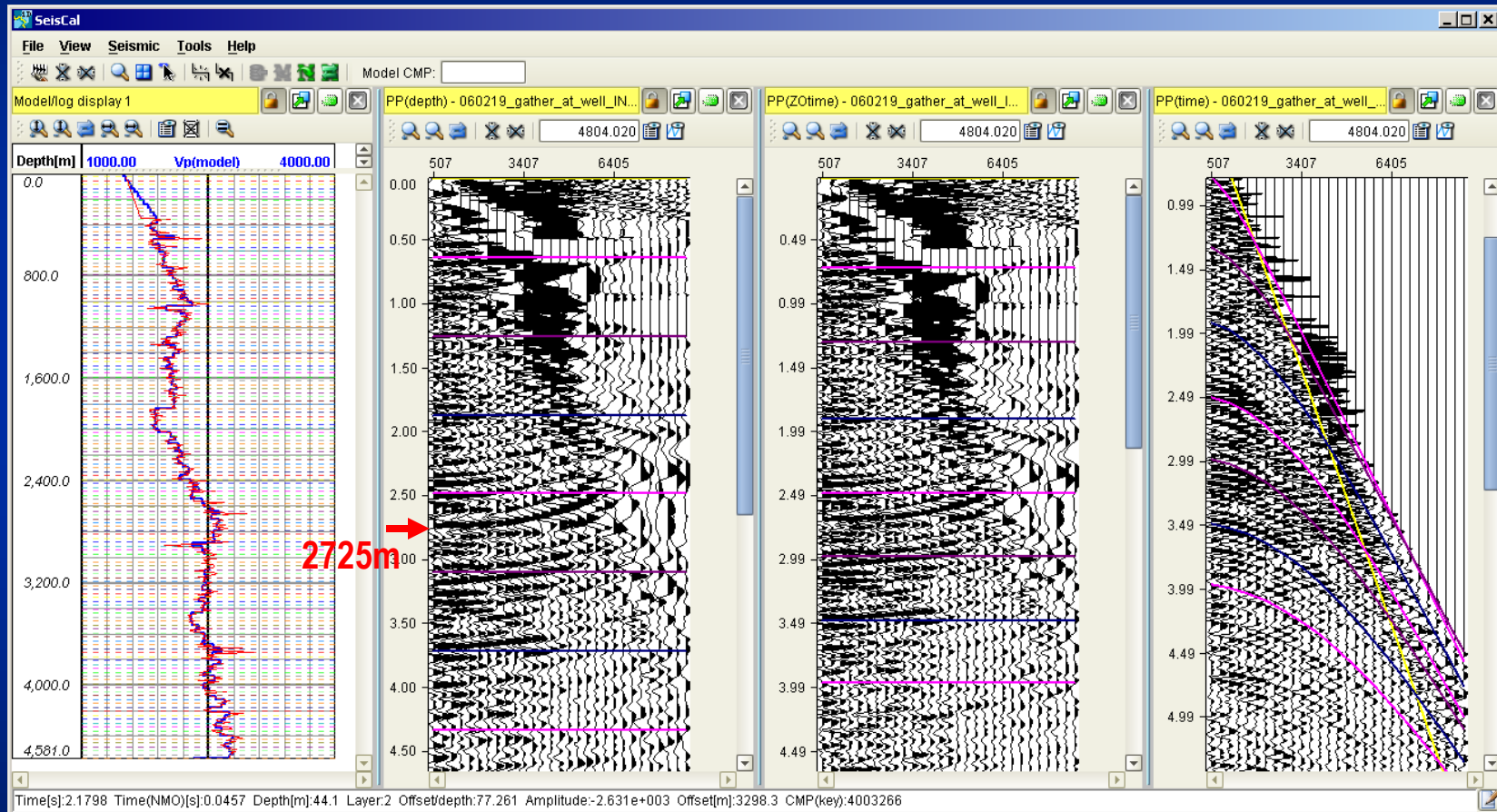
Forward modelling – Isotropic well sonic log model

Log vel / Vv model

converted depth

converted time

Input time gather



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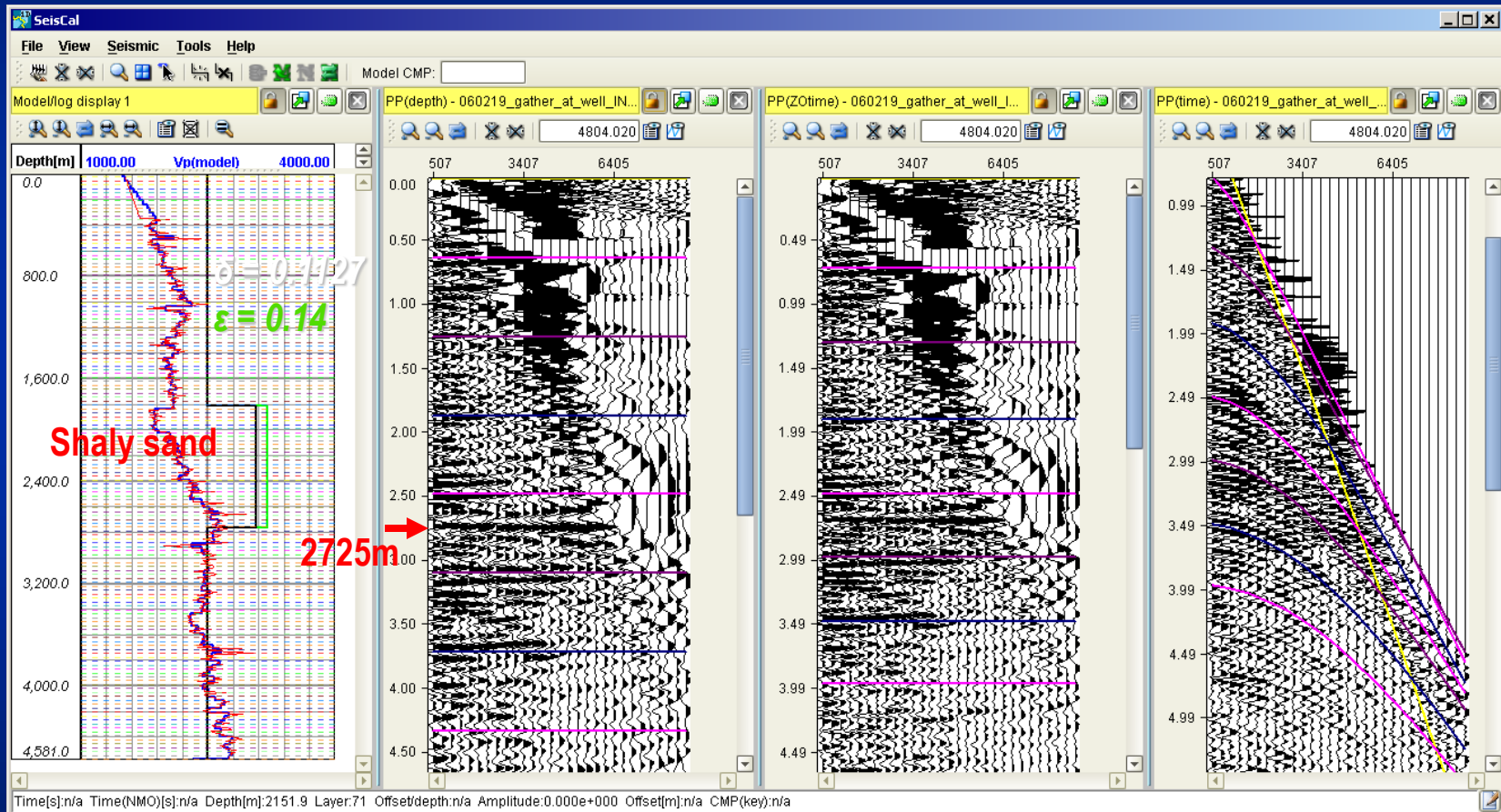
Forward modelling – VTI well sonic log model

Log vel / V_v , δ , ϵ model

converted depth

converted time

Input time gather

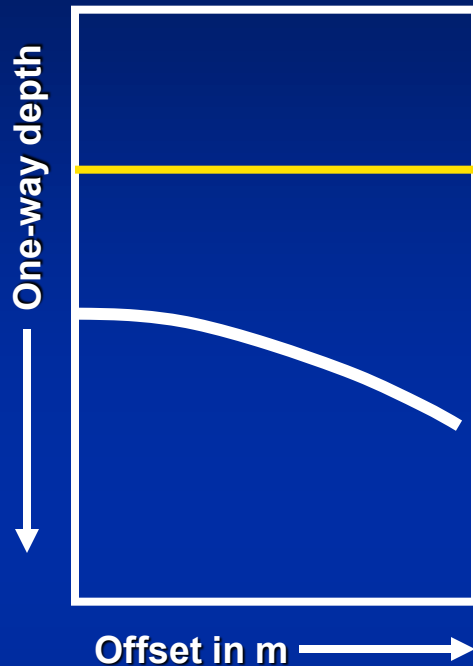


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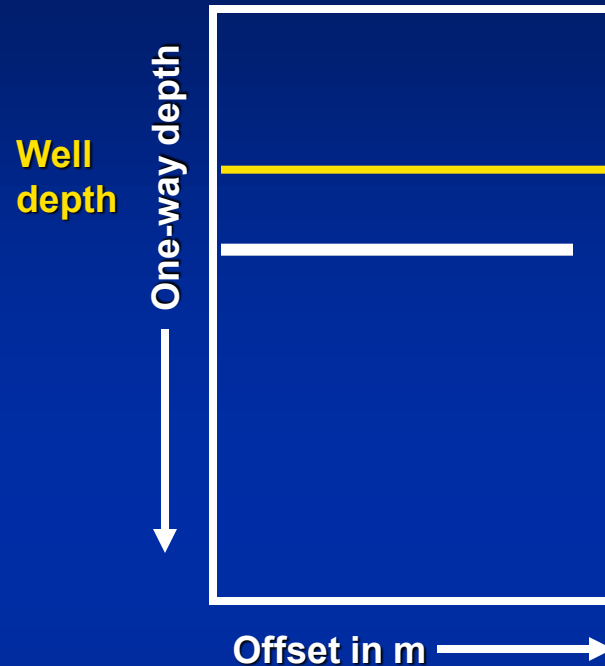
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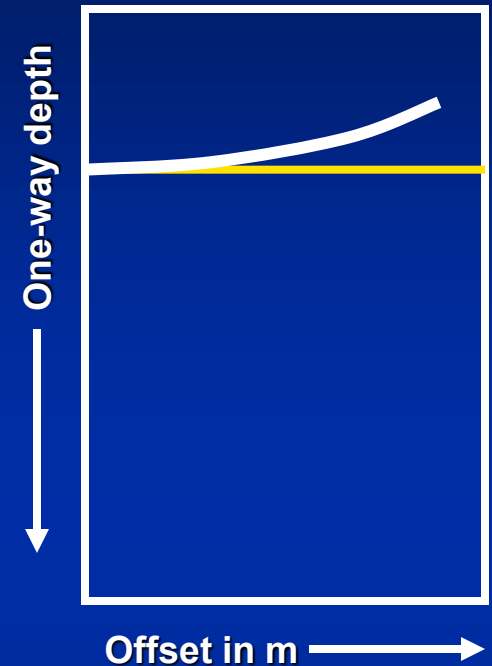
Influence of isotropic/anisotropic updates



1. Depth gather before velocity update

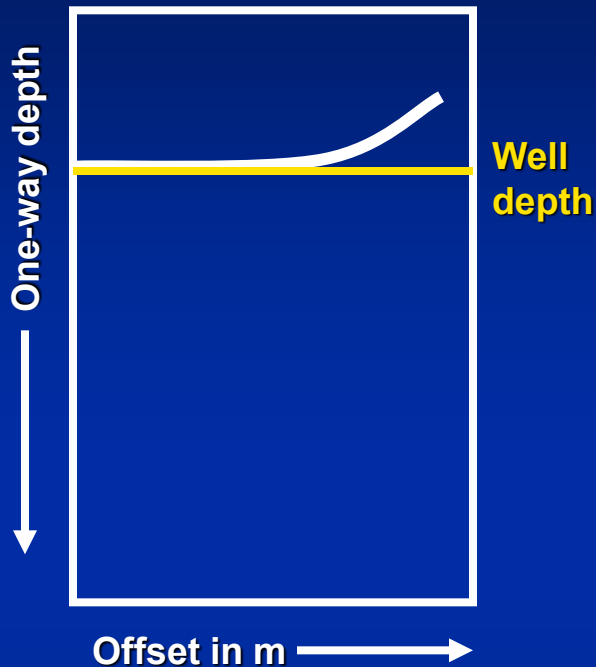


2. Depth gather after isotropic velocity update to flatten event

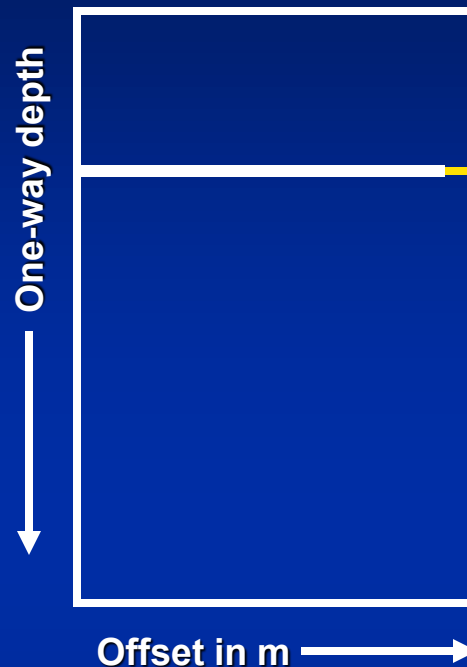


3. Depth gather after update of vertical velocity to match well depth

Influence of isotropic/anisotropic updates



4. Depth gather after update of delta to flatten near offsets



5. Depth gather after update of epsilon to flatten far offsets

Updating anisotropic parameters

- If the layer is flat then the step-by-step approach (shown on the previous 2 slides) can be used and we can 'scan' for the anisotropic parameters

BUT

- If there is dipping structure then the update of the velocity and the anisotropic parameters are all inter-related. We need to try and update all three parameters, at the same time, using tomography – **even though this is more difficult then 'scanning', it is the best approach, especially if we have a lot of well information**

Observations Regarding Anisotropy

- Anisotropic parameters slowly varying or constant
- Anisotropic Velocity Model Building and Depth Migration may be Essential for Success.
 - Improved well ties
 - More plausible velocity variations
 - More plausible structure
 - Better imaging
 - Consistency with VSP

What to Remember

- The causes of anisotropy
- The effects of anisotropy on seismic data
- Methods of solving for anisotropy

Golden rules for anisotropic model building

- All available data should be integrated into the model building:
 - well data - *Inspect sonic logs for evidence of fine layering, gamma logs – shales....*
 - more “exotic” borehole measurements – sonic scanner, RT scanner ...
 - check shots & WAVSPs
 - Geology and lithology
 - *Compaction driven basin or “hard geology” regime*
 - structural interpretation availability
 - surface seismic!!!
 - Anything else – e.g. geomechanical modelling, rock physics data, pore-pressure info
- Use knowledge of geology / geophysics to make a priori assumptions in areas where other data is inadequate
- Make the model **only as complex as the data requires**
- The model should be geologically reasonable
 - **Involve the interpreter as much as possible**

More “rules” - estimating anisotropy and running Anisotropic Tomo updates

- For Anisotropic tomographic updates make the mute more open to include the long offset terms in the picking ($>30^\circ$)
 - except for a few cases with ultra long offsets & post critical reflections
- Always carefully validate delta if it is derived “manually” from well mis-ties in case some long offset term has been included
 - Layers were not flat at all??
- Beware of well misties & SeisCal analysis with dipping structure
 - give more weight to misties & analysis locations from flatter structure

• The End